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United States Patent Application Publication

20250287084

Kind Code

A1

Publication Date

September 11, 2025

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IMAGE CAPTURING APPARATUS

Abstract

An image capturing apparatus according to an aspect of the present invention includes an air circulation conduit that is not communicated with an inside of a housing and is provided between a lens mount portion and an image sensor in an optical axis direction, and a metallic plate located on a facing surface of the image sensor.

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Family ID: 94733967

Appl. No.: 19/064434

Filed: February 26, 2025

Foreign Application Priority Data

JP 2024-033087

Mar. 05, 2024

Publication Classification

Int. Cl.: H04N23/52 (20230101); H04N23/51 (20230101); H05K7/20 (20060101)

U.S. Cl.:

CPC H04N23/52 (20230101); H04N23/51 (20230101); H05K7/2039 (20130101);

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates in particular to a heat dissipation structure for an image capturing apparatus.

Description of the Related Art

[0002] With the recent advancement in semiconductor technology, image capturing apparatuses have significantly improved in performance. The improvement in the performance of processors and imaging sensors makes it possible to read out a larger number of pixels at a higher speed.

[0003] Under such circumstances, one of the issues to be solved is heat generated from semiconductor elements. One of the important technical issues is to more effectively dissipate and release heat to elicit high performance so as to prevent occurrence of a failure due to heat and ensure the safety of every user that touches an electronic device, while exerting the high performance.

[0004] Japanese Patent Application Laid-Open Publication No. 2008-098924 discusses an image capturing apparatus having a structure in which a heat-releasing air duct that is not communicated with the inside of the image capturing apparatus is provided between an image sensor and a circuit board as a heat-generating element and a portion located at a side of the image sensor is formed of a material having a thermal conductivity lower than that of a portion located at a side of the circuit board.

[0005] According to this structure, an improvement in heat dissipation efficiency can be achieved while an adverse effect of heat transferred to an outer casing to be gripped by a user and an adverse effect of heat generated from the circuit relative to the image sensor can be prevented.

[0006] Japanese Patent Application Laid-Open Publication No. 2019-219458 discusses an image capturing apparatus having a structure in which a first housing unit including an opening formed on the back surface thereof is thermally connected to an electronic device while the opening is covered.

[0007] The image capturing apparatus further includes a second housing unit including a heat dissipation portion that is opposed to an electronic device, and the heat dissipation portion is not communicated with the inside of the image capturing apparatus and allows inflow of outside air.

SUMMARY OF THE INVENTION

[0008] According to an aspect of the present invention, an image capturing apparatus includes an image sensor, a lens mount portion to which an interchangeable image capturing lens is mountable, a housing including an opening functioning as an optical path during image capturing, a metallic plate formed on an imaging plane side relative to the housing, and an air circulation conduit formed of the housing and the metallic plate and including an intake and exhaust port, the air circulation conduit being not communicated with an inside of the housing, wherein the air circulation conduit is provided between the lens mount portion and the image sensor in an optical axis direction, and the metallic plate is located on a facing surface of the image sensor.

[0009] Further features of the present invention will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIGS. 1A to 1C are external perspective views each illustrating an image capturing apparatus according to a first embodiment.

[0011] FIG. 2 is a block diagram illustrating an electric configuration of the image capturing apparatus according to the first embodiment.

[0012] FIG. 3 is an exploded perspective view of the image capturing apparatus according to the first embodiment.

[0013] FIG. 4 illustrates details of an air circulation conduit of the image capturing apparatus

according to the first embodiment.

[0014] FIGS. 5A and 5B are sectional views of the image capturing apparatus according to the first embodiment.

[0015] FIG. 6 illustrates details of an air circulation conduit of an image capturing apparatus according to a second embodiment.

[0016] FIG. 7 is a schematic view illustrating an image capturing apparatus according to a third embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0017] The conventional structures discussed in Japanese Patent Application Laid-Open Publication Nos. 2008-098924 and 2019-219458 described above are each provided with a heat-releasing air duct in such a manner that the heat-releasing air duct is superimposed on a circuit board, which leads to an increase in the thickness of the image capturing apparatus.

[0018] In addition, in the structures described above, the housing is divided into parts, which may lead to deterioration in assembling workability and complexity of internal wiring.

[0019] Accordingly, the present invention is directed to providing an interchangeable-lens image capturing apparatus capable of effectively releasing heat from an image sensor while preventing an increase in the size of the image capturing apparatus in an optical axis direction.

[0020] Embodiments of the present invention will be described in detail below with reference to the attached drawings. In the embodiments, the same elements are denoted by the same reference numerals.

[0021] The embodiments to be described below are examples used in understanding how to implement the present invention. The embodiments described below may be altered or changed as appropriate, within the scope of the invention, depending on the configuration of an apparatus to which the present invention is applied. The present invention is not limited solely to the following embodiments.

[0022] One or more of the features or elements of the embodiments to be described below may be combined where appropriate. That is to say each of the embodiment and configurations described throughout can be implemented solely or as a combination of a plurality of embodiments or one or more features therefore where necessary or where the combination of elements or one or more features from individual embodiments in a single embodiment are beneficial.

[0023] First, a basic configuration of an image capturing apparatus **100** according to a first embodiment of the present invention will be described with reference to FIGS. 1A to 1C. FIG. 1A is an external perspective view of the image capturing apparatus **100** as viewed from the front surface thereof. FIG. 1B is an external perspective view of the image capturing apparatus **100** as viewed from the back surface thereof. FIG. 1C is an external perspective view illustrating a state where an interchangeable lens **90** is attached to the image capturing apparatus **100**.

(Basic Configuration of Image Capturing Apparatus **100**)

[0024] It is assumed that the image capturing apparatus **100** is an interchangeable-lens camera for video streaming and is used in a state where the image capturing apparatus **100** is fixed to a tripod or a support. However, the image capturing apparatus may be any other image capturing apparatus with or without an interchangeable-lens, i.e., an image capturing apparatus with a built-in or non-interchangeable lens, which can be used for image capture, video capture, video stream or the like.

[0025] The image capturing apparatus **100** does not include recording media. The image capturing apparatus **100** transmits image capturing data to a personal computer (PC), and the PC performs streaming and recording processing. The image capturing apparatus **100** does not include batteries, and thus is driven by power fed through a connected wire.

[0026] The image capturing apparatus **100** includes a lens mount **10** for mounting an image sensor **50** and the interchangeable lens **90**, and lens communication terminals **12** for communicating with the interchangeable lens **90** are provided in the lens mount **10**.

[0027] The image sensor **50** includes a mounting board on which the image sensor is mounted.

That is to say, the image sensor **50** may be considered to comprise a mounting board and a sensor for imaging, i.e. the image sensor, mounted to the mounting board. An optical filter and the like are formed on an object side of an imaging plane of the image sensor mounted on the mounting board.

[0028] An optical axis is parallel to a Z-axis direction, and light that has transmitted through the interchangeable lens **90** forms an image on the image sensor **50**, i.e. the image sensor mounted to the mounting board, thereby making it possible to capture images and moving images.

[0029] The image capturing apparatus **100** includes a front-side exterior **20** and a back-surface exterior **40** as external components. The front-side and read-side exteriors may be enclosures components. A lens unlock button **11** is provided on a side surface of the image capturing apparatus **100**.

[0030] When the lens unlock button **11** is allowed to slide to a negative side in the Z-axis direction, the lens mount **10** is unlocked, so that the interchangeable lens **90** can be mounted or dismounted. Other embodiments may use other means or the same means moving in different directions to release the interchangeable-lens.

[0031] Each of a first tripod mount **23** and a second tripod mount **24** is provided on a surface different from the surface on which the lens unlock button **11** is provided. Other embodiments may be provided with one or more than one tripod mount, and the location of the one or more than one tripod mount may be different to that described in the first embodiment.

[0032] General tripods or camera fixation accessories can be mounted on the first tripod mount **23** and the second tripod mount **24** by screwing.

[0033] If the first tripod mount **23** is fixed at a position corresponding to the lower surface of the image capturing apparatus **100**, the image capturing apparatus **100** can capture a video image in which the long side of the image sensor **50** is horizontal, thereby making it possible to capture a video image suitable for viewing on the landscape-oriented lower surface of a display of a PC or the like.

[0034] If the second tripod mount **24** is fixed at a position corresponding to the lower surface of the image capturing apparatus **100**, the image capturing apparatus **100** can capture a video image in which the long side of the image sensor **50** is vertical, thereby making it possible to capture a video image suitable for viewing on a portrait-oriented screen of a smartphone or the like.

[0035] The provision of the first tripod mount **23** and the second tripod mount **24** on the two surfaces enables the image capturing apparatus **100** to cover various fixation methods and image capturing directions.

[0036] The back surface of the image capturing apparatus **100** is provided with a communication terminal unit **41** used to establish communication and feed power.

[0037] The communication terminal unit **41** includes a universal serial bus (USB) terminal **42** and a high-definition multimedia interface (HDMI) terminal **43**. The communication terminal unit **41** connects the image capturing apparatus **100** to a switcher of a PC or a camera, a recording device, or the like using various cables, thereby enabling the image capturing apparatus **100** to be used as a web camera or a camera for video streaming.

[0038] The USB terminal **42** also functions as a power feeding terminal and is connected to an alternating-current (AC) adaptor or a mobile battery to supply power for driving the image capturing apparatus **100**.

[0039] Depending on the specifications of a connection destination, only one cable may be used to establish communication and feed power. The USB terminal **42** and the HDMI terminal **43** are merely examples of embodiments. For example, any other communication terminal such as a serial digital interface (SDI) may be used.

[0040] The image capturing apparatus **100** is not provided with a power switch, and starts output of a video image at the start of power feeding.

[0041] A first intake and exhaust port **21** and a second intake and exhaust port **22** are provided on

two respective surfaces on which the first tripod mount **23** and the second tripod mount **24** are not provided. The first intake and exhaust port may be referred to as a first cooling port. The second intake and exhaust port may be referred to as a second cooling port. The ports **21**, **22** are airflow ports or means that allow for air to be taken in or exhaust out from the image capturing apparatus. The ports **21**, **22** may also be considered heat release ports or heat release means.

[0042] The first intake and exhaust port **21** and the second intake and exhaust port **22** are intake and exhaust ports that are mainly used to release heat generated from the image sensor **50**. The first intake and exhaust port **21** and the second intake and exhaust port **22** will be described in detail below.

[0043] The provision of the first intake and exhaust port **21** and the second intake and exhaust port **22** on the two surfaces different from the surfaces on which the tripod mounts are provided reduces the risk of closing the intake and exhaust ports even when the image capturing apparatus **100** is fixed to a tripod (not illustrated). Preferably, the ports may be disposed on surfaces opposite the surface to which the mounting means are disposed.

[0044] The structure in which one of the first intake and exhaust port **21** and the second intake and exhaust port **22** is located at the position corresponding to the upper surface even when one of the first tripod mount **23** and the second tripod mount **24** is fixed at the position corresponding to the lower surface facilitates release of an upward air current due to generated heat.

(Block Diagram)

[0045] Next, an electric configuration and operation of the image capturing apparatus **100** according to an embodiment of the present invention will be described with reference to FIG. 2. FIG. 2 illustrates a block diagram of the image capturing apparatus **100** and also illustrates a main electric configuration of the image capturing apparatus **100** as a schematic view.

[0046] An optical lens unit **91** is incorporated in the interchangeable lens **90**. Although schematically illustrated in FIG. 2, in practice, the optical lens unit **91** is a lens unit including a plurality of lenses.

[0047] An autofocus (AF) drive circuit **93** causes some of the lenses in the optical lens unit **91** to advance or recede to thereby perform a focus adjustment.

[0048] An electromagnetic diaphragm **92** for adjusting the amount of light is located on an optical axis **1000** of the optical lens unit **91**.

[0049] An opening diameter of the electromagnetic diaphragm **92** can be changed by an aperture drive circuit **94** based on a designated aperture value.

[0050] The interchangeable lens **90** and the image capturing apparatus **100** are electrically connected via the lens communication terminals **12**. Based on communication with the image capturing apparatus **100**, a lens control circuit **95** controls the AF drive circuit **93** and the aperture drive circuit **94** to drive the interchangeable lens **90** according to a desired setting.

[0051] A system control unit **71** performs a main operation control operation for the image capturing apparatus **100**.

[0052] The system control unit **71** is a small micro processing unit (MPU) mounted on a control board **70**, and components of the system control unit **71** are electrically connected.

[0053] A light beam that has transmitted through the interchangeable lens **90** forms an image on the image sensor **50**, and an analog-to-digital (A/D) converter **52** extracts the output of the image sensor **50** as a digital signal.

[0054] After that, an image processing unit **53** performs processing such as filter processing and data compression processing on the digital signal, and transmits the digital signal to the system control unit **71**.

[0055] The memory control unit **72** may store image capturing results in a memory **73** depending on image processing or specifications for communication with an external device.

[0056] The image sensor **50** also functions as an automatic exposure (AE) sensor **54** and a focus detection unit **55**.

[0057] For example, the AE sensor **54** calculates exposure information based on image capturing results, and the system control unit **71** that has received the exposure information performs processing for adjusting the aperture value of the interchangeable lens **90**, the International Organization for Standardization (ISO) sensitivity of the image sensor **50**, a shutter speed, and the like so as to obtain a desired exposure.

[0058] The focus detection unit **55** has a function for calculating a driving direction and a movement amount of a focus lens based on a defocus state of an image formed on the image sensor **50** and feeding back calculation results to the AF drive circuit **93** of the interchangeable lens **90**, thereby making it possible to rapidly focus on an object.

[0059] A communication control unit **75** is connected to the USB terminal **42** and the HDMI terminal **43** to perform communication processing with an external device.

[0060] A connection to a PC or a tablet terminal makes it possible to output a captured video image and perform video image recording and streaming.

[0061] An input from an external device makes it possible to make image capturing settings for the image capturing apparatus **100**. The USB terminal **42** is also connected to a power supply control unit **74**, and can also be used as a terminal that receives power supplied from an AC adaptor of a USB output or the like.

[0062] The image sensor **50** is provided with a temperature sensor **76**. The temperature sensor **76** sends temperature information to the system control unit **71** to thereby monitor the heat generated from the image sensor **50**.

[0063] To avoid a failure due to high temperature, an image recording process can be interrupted when a predetermined threshold is reached.

(Internal Configuration of Image Capturing Apparatus **100**)

[0064] Next, an internal configuration of the image capturing apparatus **100** according to an embodiment of the present invention will be described with reference to FIG. **3**. FIG. **3** is an exploded perspective view of the image capturing apparatus **100** as viewed along the Z-axis direction.

[0065] An internal housing **30** is provided on the inside of the front-side exterior **20**.

[0066] The internal housing **30** functions as a main frame, and is assembled with various components, the image sensor **50**, the exterior including the front-side exterior **20**, and the like.

[0067] The internal housing **30** is also provided with a rectangular opening **30a** that is provided in an area other than an image capturing range so as to prevent the light beam from the interchangeable lens **90** from being blocked. That is to say, the opening **30a** is disposed such that it does not interfere with the functioning of the camera, i.e. it does not interfere with the light beam from the interchangeable-lens from being received by the image sensor. Put another way, the opening is provided in the housing such that when fitted the image sensor is not obstructed by the housing, e.g. when the image sensor is fitted it is wholly visible in a plan view through the opening.

[0068] The internal housing **30** is also provided with an air circulation conduit **31** to be described in detail below. A heat transfer metallic plate **60** is provided on a surface facing the air circulation conduit **31**.

[0069] A rubber seal **61** is sandwiched between the heat transfer metallic plate **60** and the air circulation conduit **31**, so that the opening of the air circulation conduit **31** is hermetically sealed with the heat transfer metallic plate **60**.

[0070] In other words, the air circulation conduit **31** is separated from the internal space of the image capturing apparatus **100** and is not communicated with the internal space of the image capturing apparatus **100**. This prevents entry of dust or the like and adhesion of dust or the like to the image sensor **50** and the control board **70**.

[0071] A metallic material having higher thermal conductivity, such as an aluminum alloy or copper can be used as a material for the heat transfer metallic plate **60**.

[0072] The image sensor **50** is fixed to a retaining plate **51** and is opposed to the heat transfer

metallic plate **60**.

[0073] A first elastic heat transfer member **62** and a second elastic heat transfer member **63** are sandwiched between the retaining plate **51** and the heat transfer metallic plate **60**, thereby making it possible to release the heat generated from the image sensor **50** to the heat transfer metallic plate **60**.

[0074] The first elastic heat transfer member **62** and the second elastic heat transfer member **63** are located at upper and lower positions, respectively, of the image sensor **50** in a Y-axis direction, thereby preventing the temperature of the image sensor **50** from being locally decreased and preventing a temperature distribution from being biased.

[0075] The control board **70** is located on the back surface of the image sensor **50** and is electrically connected to the image sensor **50** and the lens communication terminals **12** via a flexible printed board and connectors.

[0076] Various communication terminals of the communication terminal unit **41** are mounted on the control board **70**.

[0077] The control board **70** performs an overall control operation for driving the image capturing apparatus **100**, such as a control operation for the image sensor **50**, a control operation for communication with the interchangeable lens **90**, and a control operation for communication with an external device.

[0078] The control board **70** is screwed to the internal housing **30** through the heat transfer metallic plate **60**, and four corners of the control board **70** contact the heat transfer metallic plate **60**, thereby making it possible to release the heat generated from the control board **70** to the heat transfer metallic plate **60**. Put another way, the control board is secured to the internal housing through fixation means, preferably screws, which in use extend through the heat transfer metallic plate **60**. The four corners, or a plurality discrete locations around the periphery, of the control board are in contact with the heat transfer metallic plate to allow heat generated from the control board to be transferred to the heat transfer metallic plate.

(Air Circulation Conduit **31**)

[0079] Next, the air circulation conduit **31** of the image capturing apparatus **100** according to an embodiment of the present invention will be described in detail with reference to FIG. **4** and FIGS. **5A** and **5B**.

[0080] FIG. **4** illustrates the internal housing **30** as viewed from the back surface side in a state where the air circulation conduit **31**, except for the heat transfer metallic plate **60**, the image sensor **50**, the control board **70**, and the like, is exposed. FIG. **5A** is a front view of the image capturing apparatus **100**. FIG. **5B** is a sectional view of the image capturing apparatus **100** taken along a line A-A illustrated in FIG. **5A**.

[0081] A first intake and exhaust path **31a** is a tunnel-like flow path provided in the internal housing **30** and leads to the first intake and exhaust port **21** provided above the lens unlock button **11** on the side surface.

[0082] Similarly, a second intake and exhaust path **31b** is a tunnel-like flow path provided in the internal housing **30** and leads to the second intake and exhaust port **22** provided on the upper surface. In other embodiments, the intake and exhaust ports may be disposed in any surface, and the air circulation conduit extends between both ports.

[0083] The path leading from one or each intake and exhaust port may branch off into a plurality of paths, like the second intake and exhaust path **31b**.

[0084] The air circulation conduit **31** is located to surround the four surrounding sides of the opening **30a**. For example, air flowing in from the first intake and exhaust port **21** is allowed to pass through a path as indicated by an arrow **F1** and can be exhausted through the second intake and exhaust port **22**. Put another the air circulation conduit **31** is disposed to substantially surround the four sides of the opening **30a**, i.e. the air circulation conduit **31** extends substantially along each side of the opening.

[0085] A hatched area in the air circulation conduit **31** illustrated in FIG. **4** is opened to the back surface side, and as described above with reference to FIG. **3**, the opening is hermetically sealed with the rubber seal **61** and the heat transfer metallic plate **60**.

[0086] In other words, as illustrated in FIG. **5B**, a part of the surface of the air circulation conduit **31** is formed of the heat transfer metallic plate **60**.

[0087] With the above-described configuration, heat generated in the camera is collected on the heat transfer metallic plate **60**, and the heat transfer metallic plate **60** is also exposed to the internal surface of the air circulation conduit **31**, thereby making it possible to effectively release the heat.

[0088] The air circulation conduit **31** is located around the opening **30a**, so that the effect of uniformly releasing heat generated from the image sensor **50** is obtained.

[0089] According to the related art, in interchangeable-lens cameras, such as the image capturing apparatus **100**, the optical path length from the lens mount **10** to the image sensor **50** is uniquely standardized, so that various interchangeable lenses can be used.

[0090] Many cameras that emphasize still image capturing incorporate a mechanical shutter mechanism mounted in a space between the lens mount **10** and the image sensor **50**, and the optical path length is set also in consideration of the space.

[0091] However, in general, such a mechanical shutter mechanism is not used in moving image capturing. Therefore, like in the image capturing apparatus **100** according to the present embodiment, the necessity of the shutter mechanism for a camera that mainly uses moving image capturing functions is low.

[0092] On the other hand, the image sensor **50** is always operated in a recording state during moving image capturing. Thus, in particular, heat generated from the image sensor **50** is a significant issue.

[0093] That is, according to the present embodiment, it is possible to provide a structure for effectively releasing heat generated from the image sensor **50** while preventing an increase in the size of the image capturing apparatus **100** by utilizing the space for the shutter mechanism that is not required for a moving image capturing apparatus.

[0094] The features of the first embodiment will be described below with reference to the drawings and the reference numerals.

[0095] A first set of features are now described with reference to FIG. **4** (also shown in FIGS. **6** and **7**).

[0096] The image capturing apparatus **100** includes the image sensor **50**, the lens mount **10** to which an interchangeable image capturing lens is mountable, the housing **30** including the opening **30a** functioning as an optical path during image capturing, and the metallic plate **60** formed on the image plane side relative to the housing **30**.

[0097] The image capturing apparatus **100** further includes the air circulation conduit **31** that is formed of the housing **30** and the metallic plate **60**, includes intake and exhaust ports, and is not communicated with, i.e. in communication with, the inside of the housing **30**.

[0098] The air circulation conduit **31** is provided between the lens mount **10** and the image sensor **50** in the optical axis direction, and the metallic plate **60** is located on the facing surface of the image sensor **50**.

[0099] Next, a second, preferable, feature is described with reference to FIG. **4**.

[0100] The metallic plate **60** is thermally connected to the image sensor **50**.

[0101] Next, a third, preferable, feature is described with reference to FIG. **4**.

[0102] The housing **30** includes the tripod screw hole portions (the first tripod mount **23** and the second tripod mount **24**) that are configured to fix the housing **30** to a tripod, and the first intake and exhaust port **21** and the second intake and exhaust port **22** of the air circulation conduit **31** are provided on the surfaces on which the tripod screw hole portions (the first tripod mount **23** and the second tripod mount **24**) are not present.

[0103] Next, a fourth, preferable, feature is described with reference to FIG. **4** (also disclosed in

FIGS. 6 and 7).

[0104] The opening **30a** and an opening **130a** each have a rectangular shape. The air circulation conduit **31** is provided along the four sides of the openings **30a**, and an air circulation conduit **131** is provided along the four sides of the opening **130a**.

[0105] The first embodiment described above illustrates an example where, in the image capturing apparatus **100** for video streaming, the air circulation conduit **31** is located around the opening **30a** in the internal housing **30** and leads from the first intake and exhaust port **21** provided on the side surface to the second intake and exhaust port **22** provided on the upper surface.

[0106] A second embodiment illustrates an example where intake and exhaust ports are arranged in a different manner and the flow path branches off.

[0107] A basic configuration of an image capturing apparatus according to the second embodiment is similar to that of the first embodiment, and thus detailed description with reference to the same reference numerals is omitted.

(Internal Housing **130** of Image Capturing Apparatus **110**)

[0108] FIG. **6** illustrates an internal housing **130** of an image capturing apparatus **110** according to the second embodiment as viewed from the back surface thereof.

[0109] Like in the first embodiment, the image capturing apparatus **110** has a substantially square external shape about the optical axis, and the four corner portions of the image capturing apparatus **110** have an arc shape. In other embodiments, one or more corners may have an arc or chamfered shape.

[0110] Like in the first embodiment, the image capturing apparatus **110** includes the first tripod mount **23** provided on the lower surface and the second tripod mount **24** provided on the side surface. This configuration enables the image capturing apparatus **110** to cover various fixation directions and image capturing directions. Similar to the first embodiment, the number and location of tripod mounts may alter in different embodiments.

[0111] The internal housing **130** includes the opening **130a** that is located in an area other than the image capturing range, like in the first embodiment, and the air circulation conduit **131** that is located around the opening **130a**. That is to say, the opening **30a** is disposed such that it does not interfere with the functioning of the camera, i.e. it does not interfere with the light beam from the interchangeable-lens from being received by the image sensor. Put another way, the opening is provided in the housing such that when fitted the image sensor is not obstructed by the housing, e.g. when the image sensor is fitted it is wholly visible in a plan view through the opening.

[0112] The air circulation conduit **131** is provided with a first intake and exhaust path **131a** having a tunnel shape leading to a first intake and exhaust port **121** provided on a front-side exterior **120**, and a second intake and exhaust path **131b** having a tunnel shape leading to a second intake and exhaust port **122**. That is to say that the first and second intake and exhaust paths **131a**, **131b** may be a bore, or multiple bores, having a cross-sectional shape along the or each bore.

[0113] In the second embodiment, the first intake and exhaust port **121** is provided at a location near a corner portion of the image capturing apparatus **110**. That is to say that the port may be disposed near or on a corner portion of the image capturing apparatus.

[0114] For example, if air flows in from the first intake and exhaust port **121**, as indicated by an arrow **F2**, the air passes through the first intake and exhaust path **131a** and is led to the air circulation conduit **131** located around the opening **130a**.

[0115] The air circulation conduit **131** branches off along each side at each corner portion of the opening **130a** having a rectangular shape, and merges again at an opposing corner. That is to say, the air circulation conduit may branch off at two corner portions of the opening **130a**. The air circulation conduit at the two other corners, i.e. not the branching off corners, may provide a corner in the circulation conduit.

[0116] The air passes through the second intake and exhaust path **131b** and is then exhausted through the second intake and exhaust port **122** provided at the opposing corner of the first intake

and exhaust port **121**. That is to say that the second port may be disposed near or on a corner portion of the image capturing apparatus, preferably the second port is disposed near or on a corner portion of the image capturing apparatus different to corner portion to which the first port is disposed, more preferably the second port is disposed to an opposing corner, even more preferably, the second port is disposed to a diametrically opposed corner.

[0117] The air circulation conduit **131** according to the second embodiment can be located around the opening **130a**, and the number of portions where the flow path is curved can be reduced as compared with the configuration according to the first embodiment.

[0118] In other words, the configuration according to the second embodiment enables air to pass with lower resistance, so that it can be expected that heat can be more effectively released by convection heat transfer.

[0119] Further, the formation of the first intake and exhaust port **121** having an arc shape in the vicinity of the corner portion of the image capturing apparatus **110** makes it possible to prevent the first intake and exhaust port **121** from being entirely closed even when a tripod having a width greater than the bottom surface of the image capturing apparatus **110** is mounted on the first tripod mount **23**. That is to say the first and/or second port may extend along the arcuate corner, i.e. chamfered corner, of image capturing apparatus, i.e. the port may extend along the surface of the arc of the corner.

[0120] The same advantage applies to the second intake and exhaust port **122**.

[0121] Similarly, the second intake and exhaust port **122** can be prevented from being closed when the tripod is fixed to the second tripod mount **24**. The same can be said for the first intake and exhaust port **121**.

[0122] In addition, unlike in the first embodiment, the first intake and exhaust port **121** and the second intake and exhaust port **122** are each located in the vicinity of the corner portion, so that a side surface **S1** and a side surface **S2** can be widely utilized. For example, an advantageous effect that an operation member and a display member can be easily arranged in an area other than the intake and exhaust ports can be obtained.

[0123] Features of the second embodiment will be described briefly below with reference to the drawings and the reference numerals.

[0124] Next, a fifth, preferable, feature is described with reference to FIG. 6.

[0125] The image capturing apparatus **110** has a rectangular shape as viewed along the optical axis direction, and each corner portion of the image capturing apparatus **110** has a chamfered shape or an arc shape. The first intake and exhaust port **121** and the second intake and exhaust port **122** of the air circulation conduit **131** are each provided at a corner portion.

[0126] Next, a sixth, preferable, feature is described with reference to FIG. 6 (also disclosed in FIG. 7).

[0127] The air circulation conduit **131** branches off into two directions at a corner portion of the opening **130a**. That is to say a branch-off portion of the air circulation conduit **131** may be provided at each of two corners.

[0128] The first and second embodiments described above illustrate an example where the present invention is applied to an image capturing apparatus for video streaming. In a third embodiment, an example where the present invention is applied to an image capturing apparatus of a typical type that captures an image in a state where a user grips the image capturing apparatus will be described with reference to a schematic view.

[0129] FIG. 7 is a schematic view of an image capturing apparatus **200** according to the third embodiment as viewed from the front surface thereof.

[0130] Like in the first and second embodiments, the image capturing apparatus **200** includes an image sensor **250**, a lens mount **210** for mounting an interchangeable lens (not illustrated), and a tripod mount **223** for mounting a tripod.

[0131] The image capturing apparatus **200** also includes a shutter button **201** for the user to issue

an image capturing start instruction, and a grip portion **202** for the user to grip the image capturing apparatus **200**.

[0132] An air circulation conduit **231** indicates an image of an air circulation conduit illustrated in a manner superimposed on the image capturing apparatus **200**.

[0133] In practice, like in the first and second embodiments, the air circulation conduit **231** is provided between the lens mount **210** and an image sensor **250**, and a part of the surface of the air circulation conduit **231** is formed of a metallic plate (not illustrated) that is thermally connected to the image sensor **250**. The air circulation conduit **231** is not communicated with the inside of the image capturing apparatus **200**.

[0134] A first intake and exhaust port **221** that leads to the air circulation conduit **231** is provided in the vicinity of a corner portion of a side surface on a side opposite to the grip portion **202**.

[0135] The first intake and exhaust port **221** can be prevented from being closed even in a state where a tripod is used or the user is gripping the image capturing apparatus **200**. Further, the second intake and exhaust port **222** is provided on the upper surface of the image capturing apparatus **200** and on a side more inward than the shutter button **201** so as to prevent the second intake and exhaust port **222** from being closed and prevent exhausted air from blowing to the operation member and the user's hand.

[0136] In the third embodiment, an air-cooling fan accessory **300** is attached to a side surface of the image capturing apparatus **200**. The air-cooling fan accessory **300** includes an air-cooling fan **310** and an intake port **320** so that the air can be actively guided into the first intake and exhaust port **221** by power fed from the image capturing apparatus **200**. A built-in fan or fan accessory may be an optional additional to the first and/or section embodiments.

[0137] In particular, in an image capturing mode or the like in which a large amount of heat is generated from the image sensor **250**, the air circulation conduit **231** can be utilized in combination with an external accessory such as the air-cooling fan accessory **300**.

[0138] Embodiments of the present invention have been described above. However, the present invention is not limited solely to the embodiments and can be modified or altered in various manners within the scope of the invention. Each of the embodiment and configurations described throughout the disclosure can be implemented solely or as a combination of a plurality of embodiments or one or more features therefore where necessary or where the combination of elements or one or more features from individual embodiments in a single embodiment are beneficial.

[0139] Accordingly, the configurations according to the embodiments included in the present invention are described below.

(Configuration 1)

[0140] An image capturing apparatus **100** including an image sensor **50**, a lens mount **10** to which an interchangeable image capturing lens is mountable, a housing **30** including an opening **30a** functioning as an optical path during image capturing, a metallic plate **60** formed on an imaging plane side relative to the housing **30**, and an air circulation conduit **31** that is formed of the housing **30** and the metallic plate **60**, includes intake and exhaust ports, and is not communicated with an inside of the housing **30**, in which the air circulation conduit **31** is provided between the lens mount **10** and the image sensor **50** in an optical axis direction, and the metallic plate **60** is located on a facing surface of the image sensor **50**.

(Configuration 2)

[0141] The image capturing apparatus according to Configuration 1, in which the metallic plate **60** is thermally connected to the image sensor **50**.

(Configuration 3)

[0142] The image capturing apparatus according to Configuration 1 or 2, in which the housing **30** includes a tripod screw hole portion **23** and a tripod screw hole portion **24** that are configured to fix the housing **30** to a tripod, and the intake and exhaust port **21** and the intake and exhaust port **22** of

the air circulation conduit **31** are provided on surfaces on which the tripod screw hole portion **23** and the tripod screw hole portion **24** are not present.

(Configuration 4)

[0143] The image capturing apparatus according to any one of Configurations 1 to 3, in which the image capturing apparatus **110** has a rectangular shape as viewed along an optical axis direction, each corner portion of the image capturing apparatus **110** has a chamfered shape or an arc shape, and the intake and exhaust port **121** and the intake and exhaust port **122** of the air circulation conduit **131** are each provided at a corner portion.

(Configuration 5)

[0144] The image capturing apparatus according to any one of Configurations 1 to 4, in which the openings **30a** and **130a** each have a rectangular shape, the air circulation conduit **31** is provided along four sides of the opening **30a**, and the air circulation conduit **131** is provided along four sides of the opening **130a**.

(Configuration 6)

[0145] The image capturing apparatus according to any one of Configurations 1 to 5, in which the air circulation conduit **131** branches off into two directions at a corner portion of the opening **130a**.

[0146] According to the configurations of the embodiments described above, it is possible to provide an interchangeable-lens image capturing apparatus capable of effectively releasing heat from the image sensor while preventing an increase in the size in the optical axis direction.

[0147] While the present invention has been described with reference to embodiments, it is to be understood that the invention is not limited to the disclosed embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0148] This application claims the benefit of Japanese Patent Application No. 2024-033087, filed Mar. 5, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image capturing apparatus comprising: an image sensor; a lens mount portion to which an interchangeable image capturing lens is mountable; a housing including an opening functioning as an optical path during image capturing; a metallic plate formed on an imaging plane side relative to the housing; and an air circulation conduit formed of the housing and the metallic plate and including an intake and exhaust port, the air circulation conduit being not communicated with an inside of the housing, wherein the air circulation conduit is provided between the lens mount portion and the image sensor in an optical axis direction, and the metallic plate is located on a facing surface of the image sensor.
2. The image capturing apparatus according to claim 1, wherein the metallic plate is thermally connected to the image sensor.
3. The image capturing apparatus according to claim 2, wherein the housing includes a tripod screw hole portion configured to fix the housing to a tripod, and the intake and exhaust port of the air circulation conduit is provided on a surface of the housing on which the tripod screw hole portion is not present.
4. The image capturing apparatus according to claim 2, wherein the image capturing apparatus has a rectangular shape as viewed along an optical axis direction, and a corner portion of the image capturing apparatus has one of a chamfered shape and an arc shape and is provided with the intake and exhaust port of the air circulation conduit.
5. The image capturing apparatus according to claim 2, wherein the opening has a rectangular shape and the air circulation conduit is provided along four sides of the opening.

6. The image capturing apparatus according to claim 5, wherein the air circulation conduit branches off into two directions at a corner portion of the opening.
